MSG No. 119.320 $I_c\bar{4}m2$ [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `-4m2`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4m2`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,4,5,6,9,10,15,16]`
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[17,18,23,24,27,28,29,30]`
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[19,20,21,22,25,26,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,7,8,19,20,21,22]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[3,4,5,6,17,18,23,24]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,10,15,16,27,28,29,30]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[11,12,13,14,25,26,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,7,8,27,28,29,30]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[3,4,5,6,25,26,31,32]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[9,10,15,16,19,20,21,22]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,13,14,17,18,23,24]`

Table 5: Wyckoff site: `8e`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,3,26,28]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,4,25,27]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[5,8,30,31]`
4	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[6,7,29,32]`
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,11,18,20]`
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,12,17,19]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[13,16,22,23]`

8	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[14,15,21,24]

Table 6: Wyckoff site: 8f, site symmetry: 2.22

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,12,19,26]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,11,20,25]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,10,17,28]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4,9,18,27]
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[5,15,24,30]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[6,16,23,29]
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[7,13,22,32]
8	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[8,14,21,31]

Table 7: Wyckoff site: 8g, site symmetry: 2mm.

No.	position	mapping
1	$[0, 0, z]$	[1,2,7,8]
2	$[0, 0, -z]$	[3,4,5,6]
3	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[9,10,15,16]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[11,12,13,14]
5	$\left[0, 0, z + \frac{1}{2}\right]$	[17,18,23,24]
6	$\left[0, 0, \frac{1}{2} - z\right]$	[19,20,21,22]
7	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[25,26,31,32]
8	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[27,28,29,30]

Table 8: Wyckoff site: 8h, site symmetry: 2mm.

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,2,7,8]
2	$\left[\frac{1}{2}, 0, -z\right]$	[3,4,5,6]
3	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[9,10,15,16]
4	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[11,12,13,14]
5	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[17,18,23,24]
6	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[19,20,21,22]
7	$\left[\frac{1}{2}, 0, z\right]$	[25,26,31,32]
8	$\left[0, \frac{1}{2}, -z\right]$	[27,28,29,30]

Table 9: Wyckoff site: 16i, site symmetry: ..2

No.	position	mapping
1	$[x, x, 0]$	[1,3]
2	$[-x, -x, 0]$	[2,4]
3	$[x, -x, 0]$	[5,8]

continued ...

Table 9

No.	position	mapping
4	$[-x,\ x,\ 0]$	[6,7]
5	$[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}]$	[9,11]
6	$[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}]$	[10,12]
7	$[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}]$	[13,16]
8	$[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}]$	[14,15]
9	$[x,\ x,\ \frac{1}{2}]$	[17,19]
10	$[-x,\ -x,\ \frac{1}{2}]$	[18,20]
11	$[x,\ -x,\ \frac{1}{2}]$	[21,24]
12	$[-x,\ x,\ \frac{1}{2}]$	[22,23]
13	$[x+\frac{1}{2},\ x+\frac{1}{2},\ 0]$	[25,27]
14	$[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0]$	[26,28]
15	$[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0]$	[29,32]
16	$[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0]$	[30,31]

Table 10: Wyckoff site: 16j, site symmetry: ..2

No.	position	mapping
1	$[x,\ x+\frac{1}{2},\ \frac{1}{4}]$	[1,11]
2	$[-x,\ \frac{1}{2}-x,\ \frac{1}{4}]$	[2,12]
3	$[x+\frac{1}{2},\ x,\ \frac{3}{4}]$	[3,9]
4	$[\frac{1}{2}-x,\ -x,\ \frac{3}{4}]$	[4,10]
5	$[x+\frac{1}{2},\ -x,\ \frac{3}{4}]$	[5,16]
6	$[\frac{1}{2}-x,\ x,\ \frac{3}{4}]$	[6,15]
7	$[-x,\ x+\frac{1}{2},\ \frac{1}{4}]$	[7,14]
8	$[x,\ \frac{1}{2}-x,\ \frac{1}{4}]$	[8,13]
9	$[x,\ x+\frac{1}{2},\ \frac{3}{4}]$	[17,27]
10	$[-x,\ \frac{1}{2}-x,\ \frac{3}{4}]$	[18,28]
11	$[x+\frac{1}{2},\ x,\ \frac{1}{4}]$	[19,25]
12	$[\frac{1}{2}-x,\ -x,\ \frac{1}{4}]$	[20,26]
13	$[x+\frac{1}{2},\ -x,\ \frac{1}{4}]$	[21,32]
14	$[\frac{1}{2}-x,\ x,\ \frac{1}{4}]$	[22,31]
15	$[-x,\ x+\frac{1}{2},\ \frac{3}{4}]$	[23,30]
16	$[x,\ \frac{1}{2}-x,\ \frac{3}{4}]$	[24,29]

Table 11: Wyckoff site: 16k, site symmetry: ..2'

No.	position	mapping
1	$[x,\ x,\ \frac{1}{4}]$	[1,19]
2	$[-x,\ -x,\ \frac{1}{4}]$	[2,20]
3	$[x,\ x,\ \frac{3}{4}]$	[3,17]
4	$[-x,\ -x,\ \frac{3}{4}]$	[4,18]
5	$[x,\ -x,\ \frac{3}{4}]$	[5,24]
6	$[-x,\ x,\ \frac{3}{4}]$	[6,23]
7	$[-x,\ x,\ \frac{1}{4}]$	[7,22]

continued ...

Table 11

No.	position	mapping
8	$\left[x,\, -x,\, \frac{1}{4}\right]$	[8,21]
9	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[9,27]
10	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[10,28]
11	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[11,25]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[12,26]
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[13,32]
14	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[14,31]
15	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[15,30]
16	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[16,29]

Table 12: Wyckoff site: 16l, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	[1,27]
2	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	[2,28]
3	$\left[x+\frac{1}{2},\, x,\, 0\right]$	[3,25]
4	$\left[\frac{1}{2}-x,\, -x,\, 0\right]$	[4,26]
5	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	[5,32]
6	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	[6,31]
7	$\left[-x,\, x+\frac{1}{2},\, 0\right]$	[7,30]
8	$\left[x,\, \frac{1}{2}-x,\, 0\right]$	[8,29]
9	$\left[x+\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[9,19]
10	$\left[\frac{1}{2}-x,\, -x,\, \frac{1}{2}\right]$	[10,20]
11	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[11,17]
12	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[12,18]
13	$\left[x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[13,24]
14	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[14,23]
15	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	[15,22]
16	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[16,21]

Table 13: Wyckoff site: 16m, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{3}{4},\, \frac{1}{4},\, z\right]$	[1,26]
2	$\left[\frac{1}{4},\, \frac{3}{4},\, z\right]$	[2,25]
3	$\left[\frac{1}{4},\, \frac{3}{4},\, -z\right]$	[3,28]
4	$\left[\frac{3}{4},\, \frac{1}{4},\, -z\right]$	[4,27]
5	$\left[\frac{1}{4},\, \frac{1}{4},\, -z\right]$	[5,30]
6	$\left[\frac{3}{4},\, \frac{3}{4},\, -z\right]$	[6,29]
7	$\left[\frac{1}{4},\, \frac{1}{4},\, z\right]$	[7,32]
8	$\left[\frac{3}{4},\, \frac{3}{4},\, z\right]$	[8,31]
9	$\left[\frac{1}{4},\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[9,18]
10	$\left[\frac{3}{4},\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[10,17]
11	$\left[\frac{3}{4},\, \frac{1}{4},\, \frac{1}{2}-z\right]$	[11,20]

continued ...

Table 13

No.	position	mapping
12	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[12,19]
13	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[13,22]
14	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[14,21]
15	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[15,24]
16	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[16,23]

Table 14: Wyckoff site: 16n, site symmetry: .m.

No.	position	mapping
1	$[0,\ y,\ z]$	[1,7]
2	$[0,\ -y,\ z]$	[2,8]
3	$[y,\ 0,\ -z]$	[3,5]
4	$[-y,\ 0,\ -z]$	[4,6]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9,15]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[10,16]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[11,13]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[12,14]
9	$\left[0,\ y,\ z+\frac{1}{2}\right]$	[17,23]
10	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	[18,24]
11	$\left[y,\ 0,\ \frac{1}{2}-z\right]$	[19,21]
12	$\left[-y,\ 0,\ \frac{1}{2}-z\right]$	[20,22]
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25,31]
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[26,32]
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[27,29]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ -z\right]$	[28,30]

Table 15: Wyckoff site: 32o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$[y,\ x,\ -z]$	[3]
4	$[-y,\ -x,\ -z]$	[4]
5	$[y,\ -x,\ -z]$	[5]
6	$[-y,\ x,\ -z]$	[6]
7	$[-x,\ y,\ z]$	[7]
8	$[x,\ -y,\ z]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[10]
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[12]
13	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[14]
15	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[15]

continued ...

Table 15

No.	position	mapping
16	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[18]
19	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[19]
20	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[20]
21	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[21]
22	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[22]
23	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[23]
24	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25]
26	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[26]
27	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[27]
28	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	[28]
29	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[29]
30	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ -z\right]$	[30]
31	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[31]
32	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[32]